

RICHMOND COUNTY AP, NC, USA

WMO: 722243

Lat: 34.891N

Lon: 79.759W

Elev: 109

StdP: 100.02

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03738

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-3.4	-16.3	0.9	-1.4	-13.5	1.2	0.1	9.3	12.7	8.3	11.1	1.4	10	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	35.2	23.8	33.9	23.3	32.7	23.0	25.8	31.2	25.3	30.3	24.9	29.6	3.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.9	27.6	24.1	19.2	27.0	23.6	18.7	26.6	80.3	31.4	78.0	30.4	76.0	29.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.6	5.5	-9.7	36.9	2.4	1.3	-11.5	37.8	-12.9	38.5	-14.3	39.3	-16.0	40.2		
			-10.4	26.7	2.4	0.9	-12.1	27.4	-13.5	27.9	-14.8	28.4	-16.5	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	6.3	8.4	11.7	17.1	21.8	25.8	27.1	26.2	23.7	17.5	11.2	8.4
	DBStd	8.49	5.71	5.31	5.33	4.31	3.44	2.40	1.84	2.44	2.85	4.42	4.64	5.25
	HDD10.0	413	142	86	44	3	0	0	0	0	2	40	95	
	HDD18.3	1547	372	279	213	72	12	0	0	0	2	69	218	309
	CDD10.0	3027	29	42	98	218	366	473	529	502	412	234	76	47
	CDD18.3	1117	1	2	8	37	119	223	271	244	164	43	3	2
	CDH23.3	10296	1	13	76	349	1027	2172	2677	2307	1330	322	17	5
	CDH26.7	4244	0	0	8	84	358	982	1219	1016	506	71	1	0
Wind	WSAvg	2.1	2.2	2.5	2.6	2.7	2.3	2.1	1.9	1.6	1.9	1.9	1.9	2.0
Precipitation	PrecAvg	1145	86	82	93	76	73	118	123	130	121	88	76	76
	PrecMax	1375	183	167	156	156	145	241	231	241	273	269	207	153
	PrecMin	782	24	36	19	22	36	20	35	29	25	0	11	25
	PrecStd	139	45	33	36	34	29	57	53	55	69	61	48	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	25.2	27.9	30.8	34.0	36.7	36.5	37.5	34.1	31.3	25.3	23.8
		MCWB	17.0	17.9	17.2	18.1	21.7	23.4	24.5	23.3	21.5	22.0	18.1	19.0
	2%	DB	19.9	22.4	25.5	29.0	32.0	34.7	35.0	34.6	32.6	28.4	22.9	21.4
		MCWB	15.4	16.2	16.7	18.3	20.9	23.3	24.3	23.8	21.7	20.8	17.0	18.2
	5%	DB	17.7	19.8	23.2	27.4	30.4	33.1	33.7	33.0	31.4	26.9	21.2	19.5
		MCWB	14.2	14.9	15.5	17.4	20.8	22.8	24.0	23.6	21.8	20.0	16.6	17.3
	10%	DB	15.7	17.5	21.0	25.3	28.7	32.0	32.4	31.8	30.0	24.9	19.3	17.4
		MCWB	12.6	12.9	14.5	17.1	20.6	22.6	23.7	23.4	21.7	19.6	15.5	15.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	19.5	19.4	22.3	24.2	25.7	26.9	26.5	25.4	24.9	20.7	20.6
		MCDB	20.9	22.5	23.6	25.7	28.3	31.7	33.4	31.9	29.6	26.8	22.4	21.8
	2%	WB	16.9	17.9	18.5	20.8	23.0	25.0	26.1	25.7	24.6	23.5	19.4	19.0
		MCDB	18.5	20.7	22.7	24.4	27.2	30.8	31.8	30.7	28.5	25.7	21.1	20.6
	5%	WB	15.3	16.2	17.3	19.9	22.4	24.4	25.5	25.2	23.9	22.2	17.9	17.5
		MCDB	17.2	18.8	21.2	24.2	26.9	30.1	30.8	29.9	27.5	24.9	19.9	18.9
	10%	WB	12.9	13.8	16.2	18.8	21.7	23.9	25.0	24.6	23.2	20.7	16.4	15.6
		MCDB	14.6	16.5	19.7	23.1	26.3	29.2	29.9	29.0	26.6	23.7	18.9	17.1
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.1	12.0	12.6	11.3	10.9	10.4	10.0	10.3	11.3	11.8	10.2
		MCDBR	12.7	13.1	14.2	14.8	13.3	12.7	11.8	11.7	12.5	12.1	12.9	10.8
	5% WB	MCWBR	8.6	7.9	6.7	5.9	4.4	3.8	3.5	3.3	3.9	5.3	7.8	7.6
		MCWBR	10.4	12.1	11.5	11.4	10.4	11.0	10.5	10.0	9.6	9.1	9.7	9.8
Clear-Sky Solar Irradiance	taub		0.313	0.324	0.359	0.401	0.462	0.488	0.538	0.511	0.427	0.386	0.334	0.311
		taud	2.509	2.473	2.405	2.279	2.158	2.125	2.009	2.071	2.299	2.381	2.486	2.551
	Ebn at Noon		895	923	912	885	832	807	764	777	831	839	861	878
		Edh at Noon	83	95	111	132	152	156	175	161	123	104	84	75
All-Sky Solar Radiation	RadAvg		2.67	3.32	4.41	5.52	6.07	6.39	6.05	5.50	4.68	3.84	2.94	2.32
	RadStd		0.15	0.38	0.33	0.39	0.49	0.45	0.38	0.26	0.40	0.48	0.24	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+118

Nomenclature: See separate page